PYCLIMA EXPLORER

Interactive Climate Data Insights

Technex '26 – Hack It Out

"TURNING RAW COMPLICATING CLIMATE DATA INTO A EASY & SIMPLE COMPELLING STORY"

TEAM MEMBERS

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Presented By **CODE SLAYER**

Challenges:

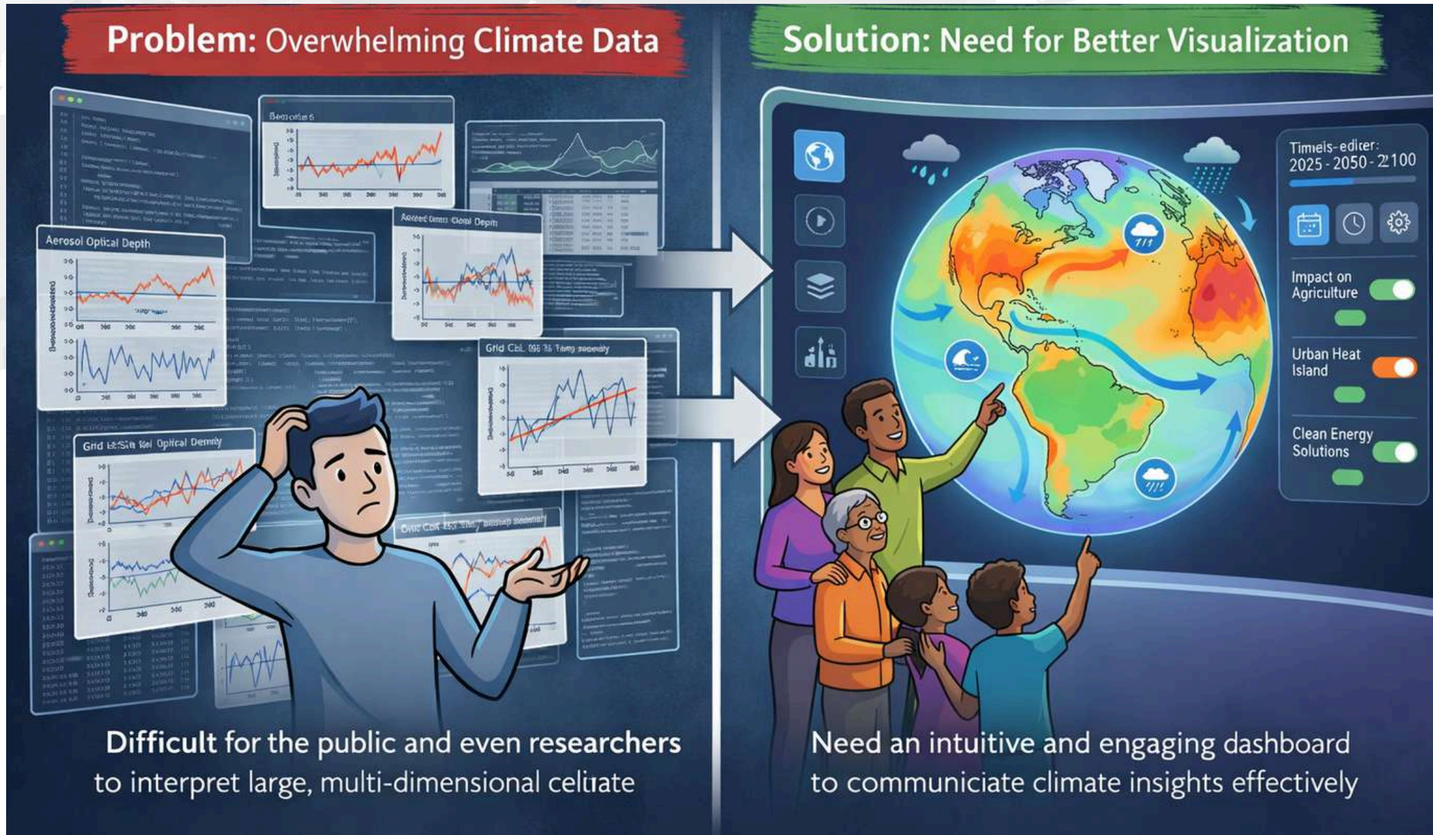
- Climate datasets stored in NetCDF are complex.
- Public and researchers struggle quick visualizations of spatial patterns and temporal trends.
- Need an intuitive interactive visualisation tool to upload, slice, and visualize variables like temperature, precipitation, wind speed.

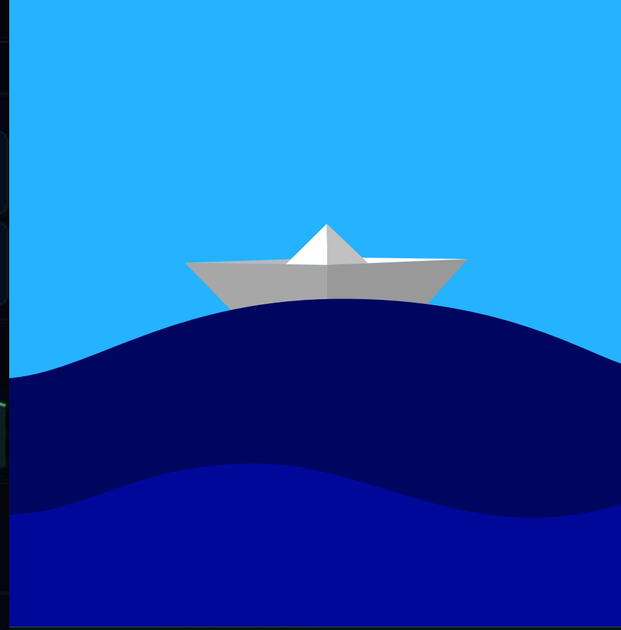
Our Vision:

- To build a web app visualiser that can transform raw numbers to a compelling stories.

Target Audience:

- Bridging the gap between the expert researchers and the climate conscious public.





Our Solution:

PyClimaExplorer — a one-stop interactive dashboard that loads NetCDF climate data, renders a 3D globe heatmap, and lets anyone explore warming anomalies in seconds.

1 — 0–3h — Data Pipeline

Download ERA5 .nc sample · xr.open_dataset() · vari extraction · Pandas flattening

2 — 3–8h — Core Viz

Plotly choropleth heatmap · px.line time series · Stre controls · variable/month selectors

3 — 8–16h — Upload & Compare

st.file_uploader() NetCDF parser · st.columns(2) split A vs Year B Δ overlay

4 — 16–22h — Bonus: 3D Globe + Story

PyDeck ScatterplotLayer globe · 4 story mode events tour · Clima Voice

5 — 22–24h — Polish & Deploy

README.md · Streamlit Cloud deploy · perf optimisation · st.cache_data · final QA

TECH STACK

USER INTERFACE

Streamlit

Climate Data Processing

Xarray | NetCDF4 | NumPy | Pandas

Deployment

Streamlit Cloud | HuggingFace | Docker

Visualisation Layer

Plotly | PyDeck | Matplotlib

AI Layer

OpenAI / LLM

FEATURES & DELIVERABLES

Spatial Map View

- Global heatmap on spinning 3D globe.
- Click any lat/lon → instant tooltip with value + anomaly reading.

Time Series View

- 12-month line chart for any pinned location.
- Dashed 1981–2010 climatology baseline for reference.

Compare Mode

- Side-by-side split-screen for any two years (e.g. 1990 vs 2020).
- Δ warming card in side panel.

Story Mode

- 4 guided anomaly events: Siberian heat wave, Arctic ice low, S.Asia monsoon surge, Sahara peak heat.

Upload .nc

- Drag-and-drop NetCDF panel.
- Auto-detects variables via Xarray.
- Supports .nc / .nc4 / .grib2.

README Docs

- Setup guide, ERA5 download links,
- Streamlit Cloud deploy steps, and data pipeline explanation.

Research Impact and Scalabilty:

Research Impact:

- Frequently analyze large datasets.
- Explore data across time, latitude, longitude and variables.
- Identifies the climate trends such as temperature trends, anamolies and extreme events.
- Useful for students, researchers and climate educators.

Scalability and Future Potentials:

- The tool is designed to handle large NetCDF files such as the 1.2 GB ERA5 datasets shown in our prototype.
- Our design using Xarray and DASK ensures the app don’t crash under pressure.
- It can be deployed on Streamlit cloud, AWS or local high performance servers.
- We’ve proven that we can transform ‘raw numbers into a compelling story’ in just few hours. This speed to market is highly attractive for companies needing quick internal data audits or public facing CSR dashboards.
- The "Story Mode" creates a "Guided Tour" framework that can be scaled to explain any specific environmental anomaly or corporate milestone.

Thank You

Let's turn raw climate numbers into a story about our changing planet

All 3 Bonus Done

- 3D Globe
- Compare Mode
- Story Mode Guided Tour